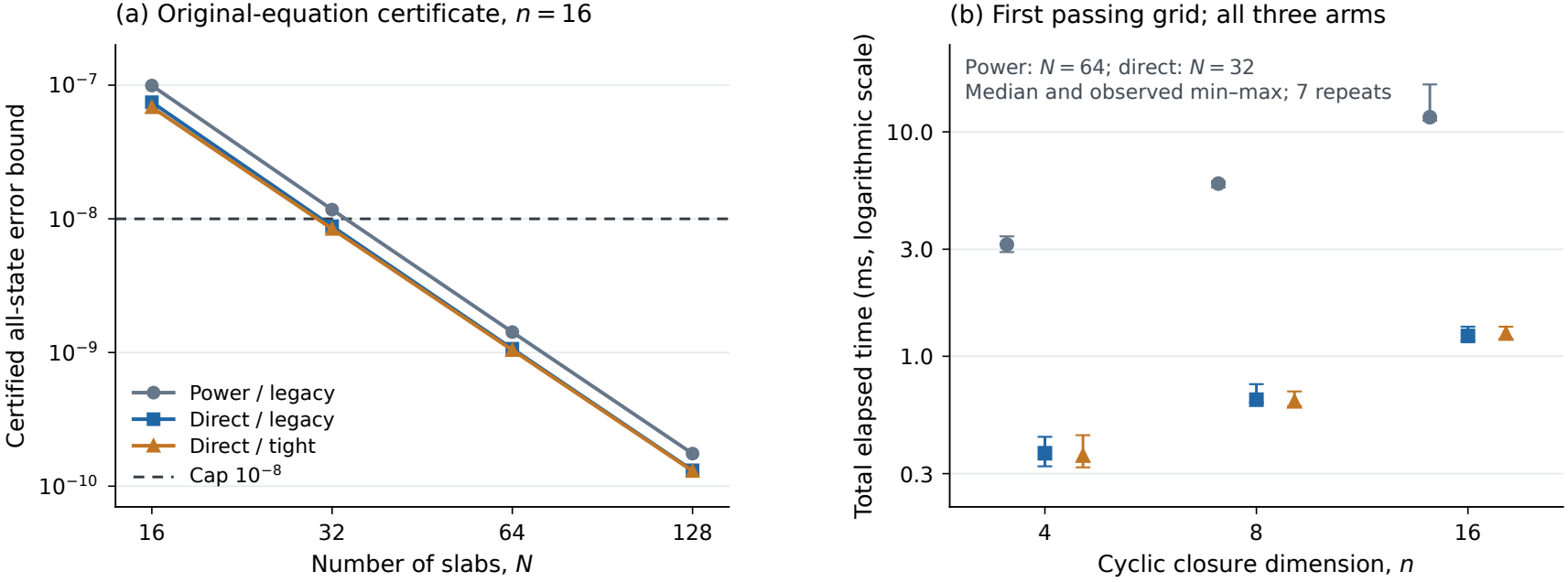
(a) Original-equation certificate, $n = 16$
Certified all-state error bound
10^{-7}
10^{-8}
10^{-9}
10^{-10}
Power / legacy
Direct / legacy
Direct / tight
Cap 10^{-8}
16
32
64
128
Number of slabs, N
(b) First passing grid; all three arms
Power: $N = 64$; direct: $N = 32$
Median and observed min–max; 7 repeats
Total elapsed time (ms, logarithmic scale)
10.0
3.0
1.0
0.3
4
8
16
Cyclic closure dimension, n
Whole call includes endpoint preparation and certification. Ranges are not confidence intervals.